

Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	29 February 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	5 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

Sairunisha S M – Team Leader

Swetha R

Lokeshwari P

Yuvashri R

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration	USN-1	As a user, I can register using my name, email, and password.	3	High	Sairunisha, Swetha
Sprint-1	User Registration	USN-2	As a user, my password will be securely encrypted before storage.	2	High	Lokeshwari
Sprint-1	Login & Authentication	USN-3	As a user, I can log in using my email and password.	3	High	Sairunisha
Sprint-1	JWT Authentication	USN-4	As a user, I can access protected APIs using a JWT token.	2	High	Yuvashri
Sprint-2	User Profile	USN-5	As a user, I can view my profile information.	2	Medium	Swetha

Sprint-2	Location Management	USN-6	As a user, I can save favorite weather locations.	3	High	Sairunisha
Sprint-2	Location Management	USN-7	As a user, I can view all saved locations.	2	High	Lokeshwari
Sprint-2	Location Management	USN-8	As a user, I can delete saved locations.	2	Medium	Yuvashri
Sprint-3	Weather Services	USN-9	As a user, I can retrieve real-time weather information for a city.	5	High	Sairunisha
Sprint-3	Weather Services	USN-10	As a user, I can receive weather details including temperature, humidity, and wind speed.	3	High	Swetha
Sprint-4	AI Integration	USN-11	As a user, I can generate AI weather summaries using Gemini AI.	5	High	Sairunisha
Sprint-4	AI Integration	USN-12	As a user, I can receive personalized weather recommendations.	5	High	Lokeshwari
Sprint-4	Testing & Deployment	USN-13	As an administrator, I can test and verify all APIs before deployment.	3	Medium	All Members

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	14 mins	24 Sep 2026	24 Sep 2026	10	24 Oct 2026
Sprint-2	09	9 mins	24 Sep 2026	24 Sep 2026	09	24 Oct 2026
Sprint-3	08	10 mins	24 Sep 2026	24 Sep 2026	08	24 Oct 2026
Sprint-4	13	13 mins	24 Sep 2026	24 Sep 2026	13	24 Oct 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\textit{sprint duration}}{\textit{velocity}} = \frac{20}{10} = 2$$

Velocity Calculation**Formula**

Velocity=Total Story Points/Sprint Duration

Calculation

- Total Story Points = 40
- Total Sprint Duration = 24 Days

Average Velocity=40/24 = 1.67

Result

Team Average Velocity = 1.67 Story Points per Day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
<https://www.atlassian.com/agile/tutorials/epics>
<https://www.atlassian.com/agile/tutorials/sprints>
<https://www.atlassian.com/agile/project-management/estimation>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

